

EAGLE EYE

[HYBRID AUTOMATIC LICENSE PLATE DETECTION & RECOGNITION USING YOLOv5 AND YOLOv8n]

**Presented
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INTRODUCTION

- ALPR is a key technology enabling Intelligent Transportation Systems (ITS) for toll collection, smart parking, traffic monitoring, and law enforcement.
- Classical image processing methods fail under blur, occlusion, lighting variations, and perspective distortion — deep learning (CNNs/YOLO) overcomes these limitations.
- YOLO-based single-stage detectors offer superior speed-accuracy trade-offs; YOLOv5 (anchor-based, CSPDarknet53) and YOLOv8n (anchor-free, C2f module) have architecturally complementary strengths.
- This project proposes a hybrid fusion pipeline: parallel YOLOv5 + YOLOv8n inference combined via Weighted Boxes Fusion (WBF), followed by a multi-stage OCR preprocessing and EasyOCR recognition module.
- Trained on 5,368 image-label pairs (NVIDIA Tesla T4 GPU); hybrid system achieves 91.67% recall — 15-19 pts higher than either standalone model — without additional training data or larger architectures.

PROBLEM STATEMENT

- Illumination Variation: Extreme lighting — sunlight glare or near-total darkness — causes detection confidence degradation in anchor-based models relying on fixed spatial features.
- Motion Blur: Fast-moving vehicles cause temporal blur degrading both detection confidence and OCR character legibility, especially in lightweight architectures.
- Perspective Distortion: Oblique camera angles cause projective distortion misaligned with axis-aligned bounding box predictions of standard YOLO detection heads.
- Precision-Recall Trade-off: Each single-model architecture has a built-in bias — optimizing precision inevitably degrades recall and vice versa — limiting performance in safety-critical deployments.
- Occlusion & Clutter: Partial blocking by vehicle parts, dirt, or environmental objects reduces detection completeness and OCR accuracy for full character sequences.

Literature Review: Vehicle Number Plate Detection & Recognition:

<u>Author(s) & Year</u>	<u>Techniques Used</u>	<u>Dataset Details</u>	<u>Result & Analysis</u>	<u>Discussion (Limitations)</u>
Aman Jain et al. (2021)	YOLOv5 (PyTorch)	Parking surveillance dataset	Efficient detection in controlled environments	Limited performance in dynamic traffic
Rayudu Sushma et al. (2022)	YOLOv5 + EasyOCR	CCTV multi-vehicle dataset	High real-time recognition accuracy	Affected by lighting and weather
Tejas Thapliyal et al. (2023)	YOLOv5 + Tesseract OCR	Multilingual plate dataset	Efficient detection and recognition	Sensitive to blur and skew
El Mehdi Ben Laoula et al. (2023)	YOLOv5 + OCR	Moroccan license plates	High precision underconditions complex	Needs retraining for other regions
Kumar Dadi (2025)	YOLOv5 Pipeline	Multi-condition vehicle images	Real-time detection achieved	Depends on preprocessing quality
B. Satya et al. (2025)	YOLOv8 Nano	Real-time ALPR dataset	Balanced speed and accuracy	Performance drops in extreme conditions
Zhang et al. (2025)	YOLOv8 Nano	General license plate dataset	Improved speed and generalization	Needs GPU for training
A. Sarhan et al. (2024)	YOLOv8 + OCR + CNN	Egyptian plate dataset	Robust detection and recognition	Limited dataset diversity
Arslan Amin et al. (2025)	YOLOv8 Nano	Custom LPR dataset	High precision and mAP	Speed-accuracy trade-off
W. Xiong et al. (2026)	Improved YOLOv8 Nano	Complex scenario dataset	Very high detection accuracy	Requires architectural tuning

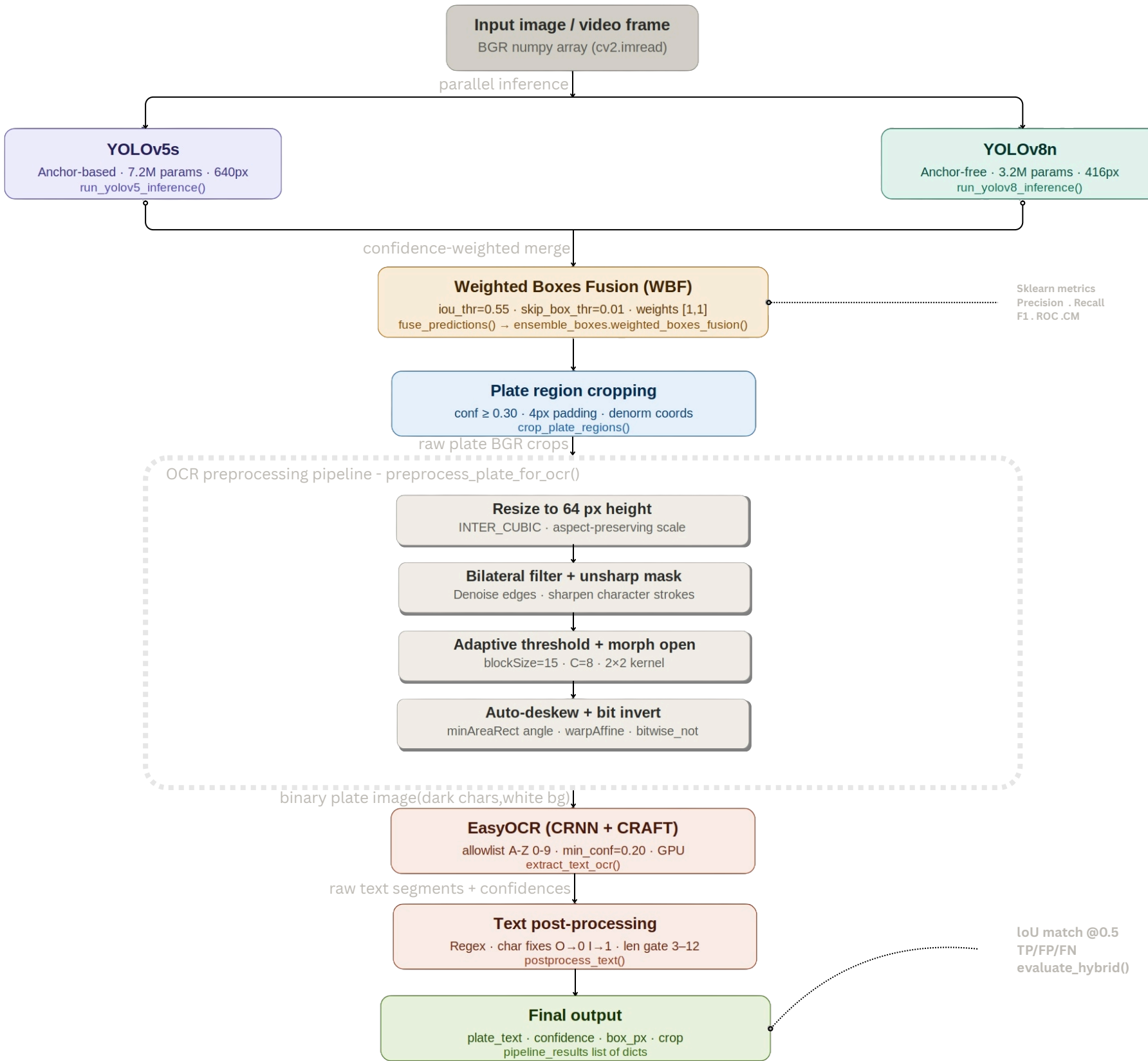
Research Gap

- Single-model architectures exhibit fundamental precision-recall trade-offs that cannot be overcome without heterogeneous model fusion
- No prior work leverages anchor-based and anchor-free YOLO detectors jointly via WBF for complementary failure mode recovery
- Existing OCR pipelines lack multi-stage preprocessing (bilateral filter + adaptive thresholding + deskewing) essential for varied plate conditions
- Recall improvement without scaling model size or training data remains unexplored in the ALPR literature
- Confidence-weighted bounding box fusion (WBF) for ALPR has not been validated on diverse real-world plate datasets with 5000+ samples
- Format-aware post-processing (O/O, I/1, Z/2 corrections) integrated with OCR confidence filtering is rarely applied in end-to-end ALPR systems

Objectives of EAGLE EYE

- Implement parallel YOLOv5 and YOLOv8n inference as complementary detection engines on the same input frames
- Apply WBF (IoU threshold 0.55) to fuse bounding box predictions with confidence-weighted averaging for geometrically refined localizations
- Develop a 5-stage OCR preprocessing pipeline (bilateral filter, sharpening, adaptive thresholding, morphological opening, deskewing) for robust character recognition
- Train both models for 50 epochs using COCO-pretrained transfer learning on 5,368 image-label pairs split 80/10/10 (train/val/test)
- Evaluate hybrid system on metrics: Precision, Recall, F1-Score, mAP@0.5, mAP@0.5:0.95, ROC-AUC, and Confusion Matrix
- Demonstrate that architecturally heterogeneous model fusion improves recall completeness without requiring larger models or additional training data

Work Flow



Legend

YOLOv5 (anchor-based)

YOLOv8n (anchor-free)

Fusion layer (WBF)

Plate cropping

OCR stage

Final output

Preprocessing sub-steps

Evaluation branch

METHODOLOGY

Step 1: Dataset Collection

- 5,368 image-label pairs from Kaggle License Plate Dataset (CC0-1.0), split 80% train / 10% val / 10% test with YOLO-format annotations.

Step 2: Model Training

- YOLOv5s (7.2M params) and YOLOv8n (3.2M params) trained 50 epochs,
- COCO-pretrained transfer learning, NVIDIA Tesla T4 GPU.

- Step 3: Hybrid Detection (WBF Fusion)
Parallel inference by both models; WBF with IoU@0.55 merges boxes,
- confidence-weighted averaging refines localization..

Step 4: Plate Cropping

- Fused bounding boxes padded and cropped from original image.
- Normalized coords converted to pixel coords for accurate crop.

Step 5: OCR Preprocessing

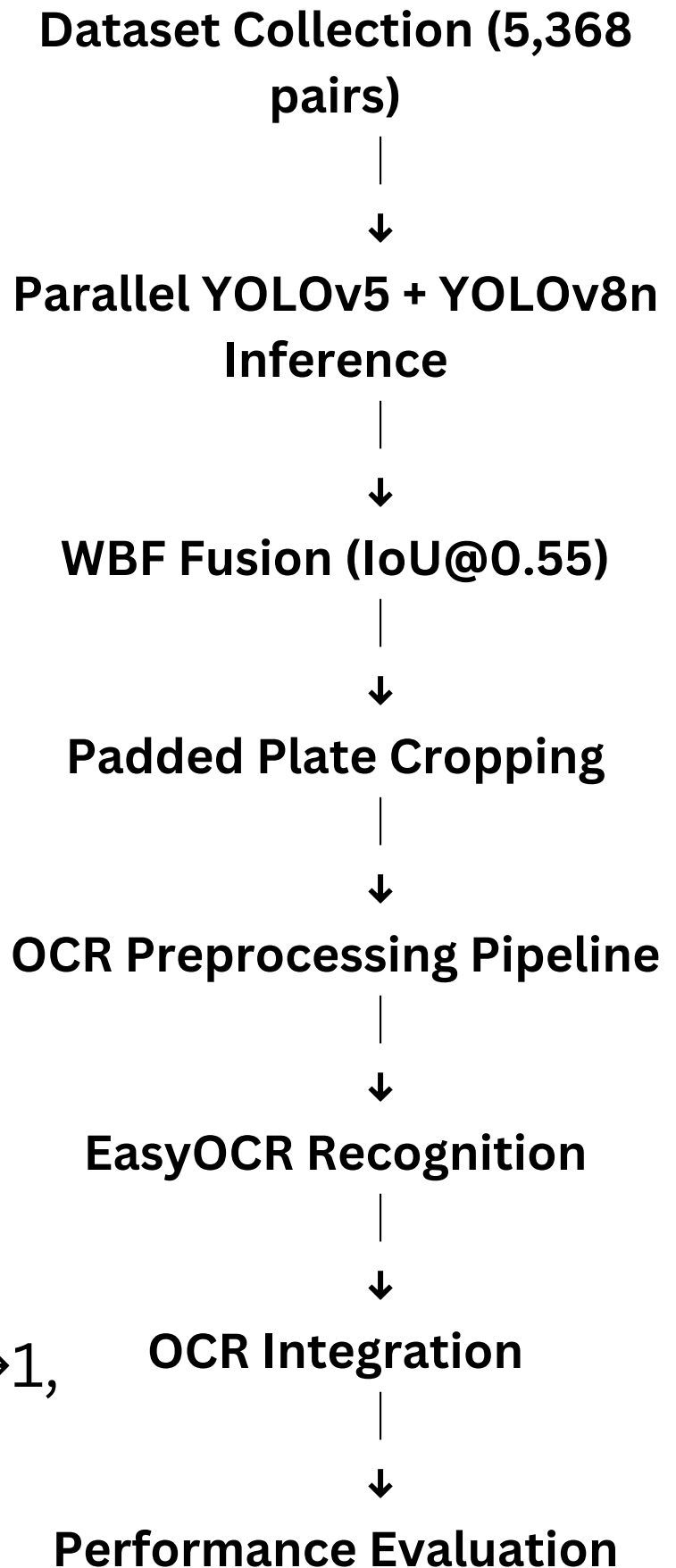
- Bilateral filter, unsharp masking, adaptive thresholding,
- morphological opening, auto-deskewing applied to each crop.

Step 6: EasyOCR Recognition

- EasyOCR with confidence threshold 0.20; char-confusion correction (O→0, l→1, Z→2, S→5, B→8); length validation (3-12 chars).

Step 7: Evaluation

- Precision, Recall, F1, mAP@0.5, ROC-AUC, Confusion Matrix.

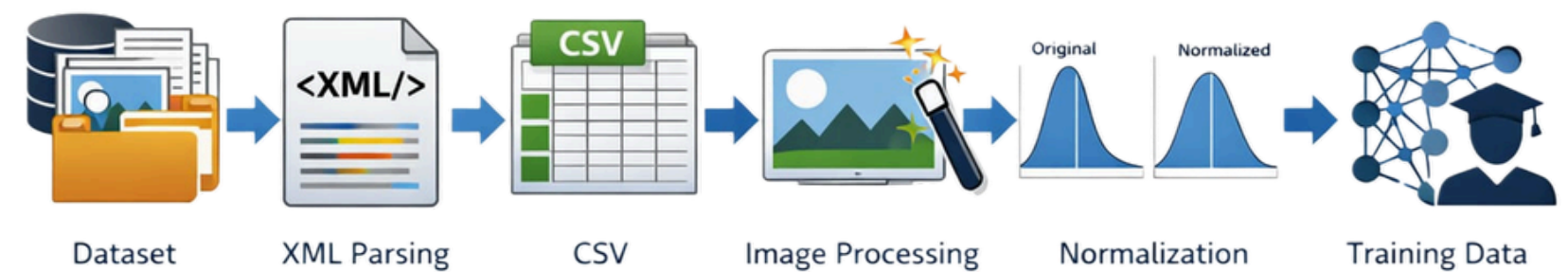


DATA PREPROCESSING

- Input: 640×640px (YOLOv5), 416×416px (YOLOv8n)
- Mosaic augmentation, random flip, HSV jitter during training
- YOLO-format normalized bbox annotations (class cx cy w h)
- Dataset: 5,368 pairs → 4,294 train / 537 val / 537 test
- Source: Kaggle License Plate Dataset (CC0-1.0 license)

MODEL TRAINING (COMPLETED)

- Epochs completed: 30
- YOLOv5s: best val mAP@0.5 ≈ 86.3%, Precision ≈ 86.2%, Recall ≈ 72.5%
- YOLOv8n: best val mAP@0.5 ≈ 78.9%, Precision ≈ 81.2%, Recall ≈ 76.7%



Result and Analysis:

- Hybrid Precision: 83.0% | Hybrid Recall: 91.67% | F1-Score: ~87.2% (50-image test subset, IoU@0.5)
- YOLOv5s standalone: Precision 86.2%, Recall 72.5%, mAP@0.5 86.3%
- YOLOv8n standalone: Precision 81.2%, Recall 76.7%, mAP@0.5 78.9%
- Recall gain of 15-19 pts over both single models — WBF recovers plates missed by one model but detected by the other
- Precision slight trade-off (expected in ensembles): more recall at cost of minor false positives
- ROC-AUC well above 0.5 baseline; confusion matrix confirms low FN rate
- OCR min confidence 0.20; char-confusion corrections improve accuracy on similar-looking characters (O/0, l/1, S/5)

FUTURE WORK

- End-to-end joint optimization: CRNN decoder attached directly to detection backbone for character-level feature alignment
- TensorRT INT8 quantization for real-time edge deployment exceeding 25 FPS on embedded GPU hardware
- Adaptive WBF weighting based on image quality (blur, illumination, scene complexity) instead of equal-weight fusion
- Temporal consistency via Kalman filtering or ByteTrack to reduce false negatives in video surveillance streams

Future Directions

- Extend to multi-line, European two-row, and Indian state-format plates
- Fine-tune EasyOCR CRNN on regional license plate fonts to reduce char-pair confusion errors
- Train/test on larger datasets: Open Images LP subset, CCPD for stronger statistical baselines
- YOLOv5s: 7.2M params, mAP 86.3%, high precision
- YOLOv8n: 3.2M params, mAP 78.9%, flexible anchor-free geometry
- Hybrid WBF: 10.4M params combined, Recall 91.67%, best F1

CONCLUSION

- Hybrid ALPR system (YOLOv5 + YOLOv8n + WBF + EasyOCR) achieves 91.67% recall — a 15-19 pt improvement over both standalone detectors — without increasing model size or training data.
- WBF fusion produces geometrically refined, tighter bounding boxes via confidence-weighted averaging, directly improving OCR crop quality and character recognition accuracy.
- Multi-stage OCR preprocessing (bilateral filter, unsharp masking, adaptive thresholding, deskewing, char-confusion correction) enables robust recognition across varied real-world imaging conditions.
- Principled use of architecturally heterogeneous model fusion exploits complementary anchor-based and anchor-free failure modes — a reusable framework for future ALPR system development.
- System demonstrated that smart ensemble fusion improves safety-critical detection completeness with affordable compute (10.4M combined parameters), suitable for real-world deployment.

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